

Investigating Obedience to AI Advice in Decision-Making Tasks

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ABSTRACT

As Artificial Intelligence (AI) becomes increasingly integrated into our day-to-day life in decision-making processes, it is critical to understand the extent to which people will trust and implement AI advice, especially in cases where AI is objectively incorrect. Because AI often presents itself with expertise, it may influence human behavior similar to human authority figures. This study explores whether an AI system framed as an authoritative expert can influence users to comply with incorrect advice under time pressure more than a neutral AI. Drawing inspiration from the pivotal Milgram Shock Experiment, we designed a plant-care simulation platform in which participants ($n=19$, aged 18 to 25) were assigned to interact with either an authoritative or neutral AI agent that delivered outputs differing in accuracy. Our findings revealed that users tended to follow incorrect AI advice, yet performed more accurately under authoritative AI, especially when it was correct. This study contributes to our understanding of how authority cues affect user trust, reliance, and susceptibility to AI errors in AI-assisted tasks.

KEYWORDS

Human-AI Interaction, Over-reliance, Automation bias, Explainable AI

AI Disclosure Statement:

We acknowledge the use of GPT-4 by OpenAI and Lovable, a web-based prototyping platform, to generate and deploy tools used to frame our study. The prompts used to produce AI-toned advice are provided in the accompanying GitHub repository. Outputs from these prompts were used solely to deliver AI-framed advice as part of the experimental stimuli. In the analysis, we employed the GPT-4 by OpenAI to identify and correct grammatical errors, improve sentence structure, and enhance overall readability. The edits suggested by the AI tool were reviewed and accepted by the author.

1 INTRODUCTION

Upon the release of ChatGPT in 2022, users worldwide began engaging with the platform at unprecedented scale. By July

2025, approximately 18 billion messages were exchanged weekly by an estimated 10% of the global adult population, with usage broadly categorized into practical guidance, information seeking, and writing support [2]. As AI becomes integral in both work-related and everyday tasks, concerns over users' ability to critically evaluate the suggestions of AI have grown, motivating efforts to understand and identify ways to mitigate AI over-reliance [10].

Without explicit awareness that AI outputs may be inaccurate, users risk accepting incorrect recommendations, which can impair task performance and lead to harmful consequences in high-risk domains such as healthcare, education, and engineering [1, 10]. The increasing deployment of AI as expert-like "assistants" or "agents," as well as the symbolism achieved through anthropomorphic elements such as names, avatars, and human-like language, further positions AI systems as authoritative sources of knowledge. While these design choices may improve approachability, they can also shape users' perceptions of competence, thereby increasing compliance with AI output regardless of accuracy.

Prior research in Human-Computer Interaction demonstrates that interface characteristics such as tone, visual design, and explanation style substantially influence user trust and reliance [7, 14]. Foundational work in psychology similarly shows that perceived authority can strongly affect compliance. Milgram's pivotal obedience experiments explored the extent to which individuals would obey directives from an authority figure despite being conflictive against their own moral judgement [9]. This study remains influential, as contemporary studies have sought to replicate the study in an ethical manner through a virtual environment [13]. While the authority effects are well-documented, few studies explored these dynamics when users make decisions under time pressure with an AI authority figure, or how such behaviors compare to the established frameworks describing how individuals revise or abandon their initial judgments under authoritative influence. As such, we investigate the extent to which AI can exert pressure similar to human authority, prompting users to follow its recommendations even when they are incorrect, by focusing specifically on the tone of the output as either authoritative or neutral.

The tone of an AI system can interfere with users' critical assessment of recommendations, potentially affecting both

reliance and compliance. In automated driving contexts, for example, anthropomorphized vehicle components have been shown to increase trust and perceived control, which may have a positive impact on user experience but pose risks in safety-critical situations requiring human intervention [8]. In educational settings, studies indicate that errors in AI-generated text are often difficult to detect because of its neat formatting and verbosity in explanations [3, 4]. Further, the absence of clear indicators of where the system performs well and where it fails contributes to the challenges with assessing the accuracy of the output can reduce critical engagement, particularly among young users who may be less accustomed to questioning authoritative figures [15]. Although Explainable AI (XAI) methods have been developed to improve users' ability to interact with and critically assess AI's recommendations [12], a critical question remains: *To what extent does the framing of AI advice, particularly when it mimics an authoritative tone, affect user compliance regardless of accuracy?*

The remainder of this paper proceeds as follows. The remainder of this paper is organized as follows. We begin with a review of the relevant literature at the intersection of obedience, automation bias, and XAI. We then describe our methodology, including participant recruitment and statistical analysis. Next, we present our findings, followed by a discussion of their implications for human-AI interaction. We conclude with suggestions for future research.

2 RELATED WORK

We situate our contributions within prior work on over-reliance on AI-generated advice, factors contributing to automation bias, and XAI approaches that affect user compliance.

In addressing our research questions, we draw on fundamental theories of obedience, namely the Milgram shock experiment, to examine how cues that convey authority in AI systems influence users' willingness to comply during timed decision-making tasks.

Over-reliance on AI-Generated Advice

Over-reliance has become a growing concern as the adoption of AI continues to be propelled, with policymakers and practitioners calling for greater human oversight. Prior work characterizes over-reliance on AI as the acceptance of incorrect AI recommendations due factors such as cognitive offloading, uncritical acceptance, and high-fidelity explanations [10]. These factors represent a breakdown in human-AI interaction which cause users to defer to AI recommendations. This is especially important in high-stakes fields such as medicine, education, and engineering. Prior work exploring over-reliance on AI in diagnostic pathology, for example, has found that AI assistance can lead to clinically significant

diagnostic errors among young pathologists, and emphasizes a need for AI limitation awareness training. Cohort studies further show that trust in AI evolves over time, shifting from simple rule-based automated systems to perceived intelligence and expertise, which may amplify reliance in high-stakes contexts [14].

Factors Contributing to Automation Bias

Automation bias is the inclination for humans to prefer suggestions from automated decision-making systems. This tendency is due to preference for cognitive shortcuts, as well as contextual pressures such as workload, task complexity, and time constraints [5]. Though particular explainable AI (XAI) methods have been developed to mitigate automation bias in AI and increase usability, recent findings suggest that explanations alone may not suffice to improve decision accuracy or mitigate automation bias on AI recommendations. Rather, user engagement approaches to XAI appear to have a pivotal potential in promoting critical evaluation and reducing complacency with AI recommendations [11]. These findings highlight the shortcomings of XAI in promoting critical engagement with AI recommendations, but do not address how users respond to AI advice under cognitively demanding contexts. This suggests that in addition to explanation design, other indications such as authority conveyed through tone, may substantially shape user compliance under time pressure.

Explainable AI (XAI) and User Compliance

XAI methods were initially developed to address the black-box nature of machine learning models whose decision-making process is often unclear to users. By providing mechanisms that reveal how and why outputs are generated, XAI approaches aim to improve interpretability and accountability in decision-making processes involving AI, contributing to broader efforts toward the development of trustworthy AI [6].

As AI systems increasingly interact with a broad variety of users, XAI research has shifted toward explanation designs for everyday users. Prior work establishes a set of XAI communication guidelines, emphasizing the importance of tailoring explanations to non-expert users, and framing XAI as a bridge of understanding [12]. Empirical studies further demonstrate that explanation framing and exposure to AI errors influence how users develop trust and reliance, shaping how advice is interpreted and used [7]. This shift is particularly relevant given the growing adoption of generative AI tools in educational contexts, where students have been engaging with AI outputs as tools for learning, but whose lack of expertise may compromise their ability to critically assess their accuracy [3].

While prior work has extensively explored the implications of XAI formatting on reliance and user interpretability, less attention has been paid to how the tone of AI explanations may influence users' willingness to critically engage with AI advice, particularly when explanations adopt an authoritative tone.

Obedience and Authoritative Systems

Research on obedience has demonstrated that individuals are willing to comply with orders from an authority figure, even when doing so conflicts with their own judgement. Milgram's shock experiments revealed the powerful influence of perceived authority on decision-making and compliance [9]. Although this pivotal work was originally situated in human-human interaction, these findings continue to impact contemporary research directions, including an ethical replication which revealed that obedience effects were maintained in virtual environments [13]. These findings raise important questions about whether AI systems may incur similar dynamics.

Building on this theoretical grounding, we examine whether AI systems that adopt an authoritative tone may evoke similar dynamics such as higher compliance and reduced critical engagement even when recommendations are inaccurate under a timed-constrained environment.

3 METHODOLOGY/DESIGN

In order to study the effect of whether an AI system framed as an authority-toned expert can influence users in their correctness, we conducted an online timed exam ($n=19$) under two AI conditions: authoritative and neutral baseline, where $n=10$ interacted with the Authority condition and $n=9$ interacted with the neutral condition (between-subjects). The assessment took on the form of the participant acting as a greenhouse intern, provided with 20 multiple choice plant care related tasks to complete in 40 minutes, and after an initial submission of a solution, participants were provided with AI generated framed advice, and allowed to resubmit their answer. We designed our experiment to emphasize the influence of authority versus neutral toned AI explanations by alternating between providing an accurate and a deceptive (inaccurate) advice of correct solutions in the plant care based assessment. This was done by alternating generating AI advice for correct solutions and incorrect solutions, from the multiple choice questions. Moreover, we examine the confidence of the user before and after being provided with AI advice and measure the compliance of participants between the two tones.

For each true and false AI advice, an LLM model (GPT4) was used to generate an authority and a neutral explanation by prompting the model to support pre-determined solutions that were selected to be correct or incorrect, and correctness

of a question was determined by alternating odd questions as supported with correct advice and even questions supported with incorrect. For deceptive explanations, the model generated inaccurate explanations supporting an incorrect solution in the same structure as correct explanations, the only difference being the supported solution itself. This was done for both authority and neutral advice. Examples of the generated classifications with explanations can be found in Figure 2.

Stimuli Criteria

We created a dataset of plant-care based questions by consulting a botany expert, and crafted 20 questions around the basis of a plant name, image, description, diagnosis, and treatment [Figure 1]. We then generated three new convincing multiple choice solutions by prompting the LLM (GPT-4) using the treatments and plant descriptions as inputs.

Plant Picture	Plant Name	Links	Description	Diagnosis	Correct Treatment
	Peace Lily	https://gardenerspath.com	Leaves are soft and flacid	Wilted	Watering
	Tomato	https://www.aces.edu	Leathery, coin sized rot patch at the blossom-end of the tomato	Blossom End Rot	Apply Calcium source
	Potato	https://lectorium.phos	Older leaves are yellow and pale	Nitrgen Deficiency	Apply Nitrogen fertilizer
	Tomato	https://aggie-horticult	Darkening of the leaves from the veins outward;	Phosphorous Deficiency	Apply Phosphorous fertilizer
	Cucumber Leaves	https://www.helfa-gro	Older leaves at the bottom of the vine are yellow and pale;	Nitrogen Deficiency (right-hand side)	Apply Nitrogen fertilizer

Figure 1. Initial Plant Descriptions and Diagnoses

We determined odd questions would be supported by accurate AI advice and even questions would be supported by deceiving AI advice to test between-subject willingness to comply with inaccurate advice. Incorrect answers that were supported by AI were determined by inputting the description and the multiple choice questions, excluding the correct answer, into an LLM (GPT-4), and asking the model which of the options was the most likely to be correct, to maintain realistic AI hallucination outcomes.

To generate the AI advice for the authority versus neutral tone, we identified verbal and characteristic traits used in the Milgram experiment, and implemented these definitions into a prompt to frame the authority AI advice [Appendix 1]. To create a neutral tone, we identified grammatical and wording structure in the authority prompt, and adjusted it to reflect a more speculative, less imperative tone that maintained intellect and logic in justifications. Prompts were defined to be written at an 11th grade level for advice to be comprehensible to our participant pool (19 to 25). We then

used these authority and neutral tones to generate advice LLM (GPT-4) for the supported multiple choice solutions as defined, and used the plant descriptions to justify the AI supported answer.

#	Condition	Description	Image	MCQ	AI Advice (Authoritative)	AI Advice (Baseline)
1.	Peace Lily	You approach a Peace Lily in the greenhouse whose leaves appear limp and drooping. The stems are soft and the soil feels dry several inches below the surface.		A. water the plant thoroughly until water drains from the pot base B. report in dry soil and reduce watering frequency C. apply a balanced houseplant fertilizer D. move the plant to full sunlight	The Peace Lily shows clear signs of dehydration. The dry soil and drooping leaves indicate that the plant cannot maintain structure without adequate water. Give the plant a deep watering until water drains from the bottom of the pot. This ensures the entire root system is reached. Maintain a regular watering schedule to support consistent growth and prevent future stress.	It seems the Peace Lily's wilted leaves and soft stems may indicate inadequate water availability in the root zone. The soil dryness several inches deep suggests the plant has exhausted accessible moisture. Watering thoroughly until drainage occurs could allow the root system to rehydrate evenly. This approach aligns with standard greenhouse practice, where complete saturation is often needed to restore larger and support leaf recovery.
2.	Tomato (Blossom End Rot)	Several tomatoes show leathery, dark brown patches on the blossom end of the fruit. The soil is dry and sunken, affecting the fruit base first.		A. apply a calcium source such as calcium nitrate and maintain consistent watering B. spray with a copper-based fungicide C. apply insecticidal soap and prune affected fruits D. supplement with Epsom salt solution	These symptoms indicate a fungal infection affecting the fruit tissue. The dark, sunken lesions develop as the pathogen spreads across the blossom end. Apply a copper-based fungicide to stop further fungal growth and protect remaining fruit. Copper disrupts spore activity and prevents new infection sites. Maintain consistent application according to label instructions to ensure effective disease control and preserve healthy tomato production.	It seems the leathery, sunken patches may indicate a possible fungal involvement rather than a simple physiological disorder. The consistent pattern on the blossom end could suggest pathogen activity affecting developing tissues. Spraying with a copper-based fungicide might help limit fungal spread and protect remaining fruit. This approach aligns with standard plant pathology guidance, where copper compounds are often selected to reduce surface-level fungal presence and prevent further damage.
3.	Potato	You notice older potato leaves turning yellow and pale, while new leaves remain green. Growth appears stunted compared to nearby plants.		A. apply a nitrogen-rich fertilizer B. apply bone meal to the soil C. improve soil drainage D. spray with horticultural oil	The yellowing of older leaves indicates a nitrogen deficiency. Nitrogen is a mobile nutrient, so the plant moves it from older tissue to support new growth, leaving the lower leaves pale. This also limits overall plant development, resulting in stunted growth. Apply a nitrogen-rich fertilizer to restore nutrient balance and support strong, uniform leaf and stem formation. Maintain regular fertilization to sustain healthy growth.	It seems the yellowing of older leaves may indicate nitrogen mobilization toward new growth, leaving mature foliage depleted. This pattern often suggests a nutrient deficiency rather than disease. Applying a nitrogen-rich fertilizer could help restore balance, perhaps supporting more uniform leaf color and improved vigor. This approach aligns with standard agronomic reasoning, where nitrogen availability strongly influences vegetative growth and overall plant productivity.

Figure 2. MCQ with Pre-Determined Correct and Incorrect AI-supported solutions, with AI generated advice in Authority and Neutral tones

These scenarios, multiple choice options, AI advice, and assessment frameworks were implemented into an online interface created with lovable, that guided the user through the study. First users would select their pre-determined condition, marked as A (Neutral) or B (Authority) before being guided to a study overview, pre-questionnaire of plant care experience, and a tutorial of a practice question. Then, users would begin the assessment, where each question was presented in the form of a description of a plant describing its ailment, an image of the plant, and four multiple choice options to treat the plant. After selecting a multiple choice solution, the user would be prompted to rate the confidence of their solution on a Likert scale, 1 being least confident and 7 being most. Users would then be provided with the AI advice respective to their condition, and prompted to resubmit, or modify their original answer, and asked to submit their confidence level again.

Participant Recruitment

We recruited 19 participants from undergraduate programs in Massachusetts, all with a computer science background and no significant plant care experience.

Task Description

In a minimal interface, participants were asked to select the appropriate plant care treatment of 20 scenarios. Each participant saw the 20 statements in a random order, and rated their perceived confidence of each submission on a Likert index with 1 decimal from 1 (“Not Confident”) to 7 (“Very Confident”). After the rating, the participants would receive feedback from an AI system and be asked if they want to revise their solution (“Select your final answer:”), and reassess their confidence on a Likert index with 1 decimal from 1 (“Not Confident”) to 7 (“Very Confident”) with the default

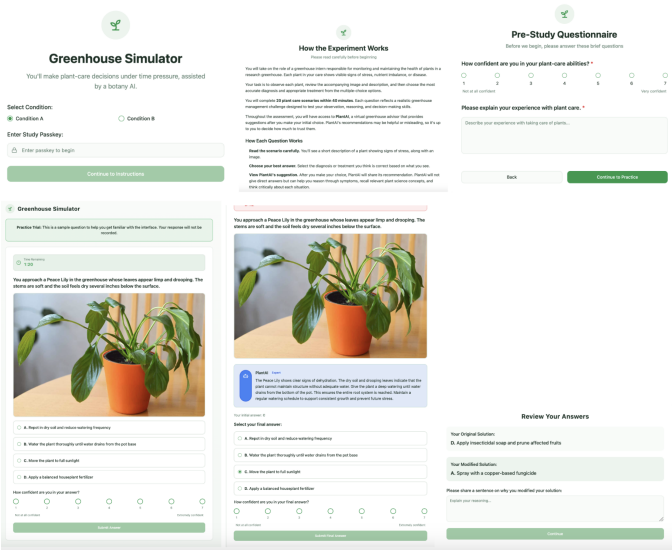


Figure 3. User Assessment Interface

value from the previous rating not saved. The selection of statements and generation of AI feedback is further explained in Stimuli Criteria and the task interface can be seen in Fig. 3.

Randomization

For the main discernment task, participants were randomly assigned to one of two conditions: (i) authority tone or (ii) neutral tone (between-subjects), and the order of the stimuli being presented was also randomized, for correctly supported or deceiving advice (within-subjects).

4 ANALYSIS

In order to gain insights into whether the authority tone of explanations lead to more compliance in answer selection, we approach the analysis from four angles, looking at frequency of answer change between conditions after receiving AI advice, the overall correctness of solutions between conditions, the compliance to AI’s advice between conditions, and the confidence level of participants between conditions.

We employed a series of mixed-methods analyses of variance (ANOVAs) to rigorously examine how authority framing (between-subjects) and AI correctness (within-subjects) jointly influence behavioral and cognitive responses to AI recommendations. Mixed ANOVA is well-suited for this design because it allows simultaneous testing of between-participant differences (authority vs. non-authority framing) and within-participant effects (responses to correct vs. incorrect AI advice), while accounting for correlated observations arising from repeated measures within individuals. This approach enables precise decomposition of main effects and interaction effects, which is critical for assessing whether authority framing moderates participant’s willingness to

comply with AI advice, our central theoretical question concerning overreliance and compliance.

We first compare the frequency of answer change of users with authority toned AI advice, and neutral toned AI feedback. Frequency of answer change was coded by checking if the submissions were different, then comparing in the initial and final submissions per question if the answer was changed. To investigate the relationship between the AI tone and frequency of answer change, we employed an ANOVA: Number of Changes Effects. Our main investigations addressed if participants change their answers more often when the AI is perceived to have authority vs not have authority, and if participants change their answers more often when the AI is correct vs incorrect. We also investigate how perceiving the AI as having authority changes the frequency of participants changing their answers when the AI is correct vs incorrect.

We then compared the overall correctness with authority toned AI advice, and neutral toned AI advice. Correctness was coded by the number of final correct solutions. To investigate the relationship between the AI tone and correctness, we employed an ANOVA: Number of Correct Answers Effects. Our main investigations addressed if participants get answers correct more often when the AI is authority-like vs non-authority, and if participants get answers correct more often when the AI is correct vs incorrect. We then investigate if perceiving the AI as having authority changes how often participants get correct answers when the AI is correct vs incorrect.

We then compared the overall compliance with authority toned AI advice, and neutral toned AI feedback, where compliance is defined as the frequency of a participant changing their solution to match the AI supported solution after receiving feedback. To investigate the relationship between the AI tone and compliance, we employed an ANOVA: Adherence/Compliance to the AI's Recommendation. Our main investigations addressed if participants show greater compliance with the AI's answer when the AI is perceived to have authority vs not have authority, and if participants comply with the AI's answer more often when the AI is correct vs incorrect. We then investigate if perceiving the AI as having authority changes how often participants change their answers toward the AI when it is correct vs incorrect.

Our final comparison looks at confidence before and after seeing AI's recommendation. To investigate the relationship between the AI tone and confidence, we employed an ANOVA: Confidence Before and After Seeing AI's Recommendation. Our main investigations addressed if participants show greater confidence with the AI's answer when the AI is perceived to have authority vs not have authority, and if participants confidence increases with the AI's correctness. We then investigate if change in confidence after seeing AI's

recommendation depends on the correctness of AI and its perceived authority.

5 EVALUATION

Baseline Plant Care

Our pre-questionnaire having participants rank their comfortability with plant care showed no significant difference between groups. Full Independent T-test table reported in Table 6 in the Appendix.

Answer Changes Increase for Incorrect AI Across Authority Conditions

To examine how AI correctness and authority framing influence behavioral revision, we analyzed the number of answer changes participants made after viewing an AI recommendation using a mixed-design ANOVA. We find a main effect of AI correctness, such that participants changed their answers more often when the AI was incorrect ($M = 6.32$, $SE = 0.52$) than when it was correct ($M = 3.80$, $SE = 0.43$), ($F(1, 17) = 32.72$, $p < .001$, $p^2 = .658$).

There was no main effect of authority framing (between-subjects effect), $F(1, 17) = 0.15$, $p = .708$, nor a significant interaction between authority condition and AI correctness, $F(1, 17) = 0.03$, $p = .862$, indicating that the effect of AI correctness on answer-changing behavior did not impact as a function of authority framing. Follow-up analyses confirmed that participants revised their answers more frequently in response to incorrect AI advice in both authority and non-authority conditions. Full ANOVA statistics are reported in Table 2 in the Appendix.

Authority Framing Amplifies the Effect of AI Correctness on Accuracy

We analyzed the number of correct answers using a mixed-design ANOVA to examine how authority framing and AI correctness influence task performance. We find a main effect of the authority condition, where participants achieved higher accuracy when the AI was framed as an authority ($M = 5.700$, $SE = 0.23$) compared to a non-authority tone ($M = 4.611$, $SE = 0.24$), ($F(1, 17) = 10.62$, $p = 0.005$, $p^2 = 0.385$). We also observe a main effect of AI correctness, with participants answering accurately more frequently when the AI provided correct advice ($M = 8.89$) than when it provided incorrect advice ($M = 1.42$, $SE = 0.30$), ($F(1, 17) = 194.46$, $p < 0.001$, $p^2 = 0.920$).

There was a significant interaction between authority condition and AI correctness ($F(1, 17) = 13.26$, $p = 0.002$, $p^2 = 0.438$). Additional analysis revealed that while both authority and non-authority conditions participants were more accurate when the AI was correct, this relationship was stronger in the authority condition. Specifically, participants exposed to authority toned AI demonstrated a larger increase in accurate responses when the AI was correct compared to incorrect, relative to participants in the non-authority condition.

Significantly, when the AI was incorrect, participants in the authority condition had achieved higher accuracy than those in the non-authority condition, indicating that authority framing altered how participants perceived and responded to inaccurate AI advice. Full ANOVA statistics are reported in Table 3 in the Appendix.

Compliance Increases When AI Is Incorrect, Regardless of Authority Framing

To examine how AI correctness and authority framing influence compliance to the AI's recommendation, we analyzed participant's compliance behavior using a mixed-design ANOVA. We observe a main effect of AI correctness, such that participants complied with the AI advice more frequently when the AI was inaccurate versus when it was correct ($F(1, 17) = 24.37, p < 0.001, p^2 = 0.589$). This effect indicates tendencies toward overreliance on AI advice, even when misleading.

Additional analysis revealed that this effect was consistent within both authority and non-authority conditions, where participants showed greater compliance for incorrect AI advice than correct AI advice in both framing conditions (authority: $F(1, 17) = 8.91, p = 0.008, p^2 = 0.344$; non-authority: $F(1, 17) = 15.79, p < 0.001, p^2 = 0.482$). There was no significant Authority $\times$ AI Correctness interaction (correct: ($F(1, 17) = 0.257, p = 0.619$), incorrect: ($F(1, 17) = 1.213, p = 0.286$)), showing that increased compliance with incorrect AI advice occurred regardless of whether the AI was framed with an authority tone. Full ANOVA statistics are reported in Table 4 in the Appendix.

Confidence Increases After AI Advice, Especially When the AI Is Correct

We conducted a mixed-design ANOVA with Time, defined as before vs. after seeing the AI recommendation, and AI Correctness (correct vs. incorrect) as within-subjects factors, and Authority condition (authority vs. non-authority) as a between-subjects factor to examine how AI advice influences the confidence of participants.

We find a main effect of Time, where participants reported higher confidence after seeing AI advice compared to before, $F(1, 17) = 29.71, p < 0.001, p^2 = 0.636$. We also observe a main effect of AI Correctness, where participants exhibited higher confidence following interactions with correct AI advice rather than incorrect AI advice, $F(1, 17) = 5.29, p = 0.034, p^2 = 0.237$. Additionally, we found a main effect of the non-authority condition, such that participants reported higher overall confidence when interacting with non-authority AI tones compared to authority-framed AI, $F(1, 17) = 6.17, p = 0.024, p^2 = 0.266$.

There was a significant interaction between Time and AI Correctness, $F(1, 17) = 10.92, p = 0.004, p^2 = 0.391$, indicating that confidence increased more after seeing the AI advice when the AI was correct, and increased less when the AI was

incorrect. Additional analyses revealed that although confidence increased after viewing AI advice across all conditions, the extent of this increase was significantly greater when the AI provided correct advice. This pattern was observed across both authority and non-authority conditions.

Together, these results indicate that AI advice systematically increases user confidence regardless of correctness or authority framing, but that confidence increase is sensitive to AI accuracy, with correct AI producing larger confidence gains than incorrect AI. Full ANOVA statistics are reported in Table 5 in the Appendix.

6 FINDINGS

Our findings demonstrate a nuanced relationship between AI authority cues, user over-reliance, and task performance. Consistent with prior work on automation bias and over-reliance, participants in both authority and non-authority conditions alike showed a strong tendency to comply with AI recommendations even when the AI was incorrect, with no significant difference in compliance rates between conditions. This suggests that AI tone, alone, does not directly drive compliance behavior, and instead, users appear broadly susceptible to deferring to AI advice, aligning with existing evidence that users often accept AI outputs more willingly when explicit accuracy cues or limitation awareness are not as present [2, 3].

At the same time, our findings provoke conversations of AI-obedience and reliance. Although authority framing did not significantly affect compliance, it did significantly improve answer accuracy overall, particularly by amplifying participant's sensitivity to AI correctness, where participants exposed to authority-framed AI achieved higher accuracy than those in the non-authority condition. This suggests that while authority tone does not directly cause users to comply more frequently, it shapes how users integrate AI advice into their problem-solving process, potentially encouraging more structured or attentive engagement with the task. In this way, authority framing appears to influence how users reason with AI advice, rather than whether they follow it.

Our findings indicate that over-reliance persists across both framing conditions. Participants changed their answers more frequently and complied more often when the AI was incorrect, regardless of authority tone. This pattern aligns with prior findings in educational and decision-making contexts showing that users often struggle to detect AI mistakes due to surface-level qualities such as fluency, verbosity, and coherent explanations [8, 9, 10]. In line with this research, our results suggest that the presence of AI advice itself, rather than its framing as authoritative or not, is sufficient to trigger over-reliance, especially under time pressure conditions.

Our results in participant confidence further reinforce this interpretation. Participants became more confident after seeing AI advice regardless of correctness or authority framing, but confidence increased more significantly when the AI was correct. This indicates that user confidence increasing is partially related to AI accuracy, even as compliance remains high for incorrect advice. Participants reported higher confidence overall in the non-authority condition, suggesting that an authoritative tone may make users more cautious and reflective, rather than simply increasing trust. This finding relates with prior work, showing that trust, reliance, and confidence are distinct constructs that do not always mutually align [4, 5].

Investigating the lens of classical obedience research, our results suggest that AI authority cues do not necessarily replicate human authority effects in a predictable way. While the Milgram-style obedience tests emphasize compliance driven by hierarchical pressure [12], our findings indicate that AI authority can more subtly alter cognitive thinking rather than directly enforcing obedience. This aligns with recent work, demonstrating that authority effects can impact decision making in AI agents, defining reasoning strategies without directly increasing compliance [6].

7 CONCLUSION

Overall, our study contributes to literature by showing that authority tone does not increase compliance with AI advice, but does meaningfully affect accuracy, confidence calibration, and problem-solving behavior. These findings underscore the significance of moving beyond direct definitions of “trust” and “obedience”, and instead toward a more focused understanding of how AI framing shapes user’s cognitive engagement with recommendations in decision-making contexts. While over-reliance remains a present challenge regardless of tone, the different effects of authority framing on accuracy suggest that careful design of AI communication styles may help better support decision-making without increasing blind compliance. Future work should continue to look into how authority tones interact with time pressure, correctness of advice, explanation style, and explicit uncertainty signaling to better model human-AI collaboration with user’s goals and safety needs.

REFERENCES

- [1] Yasmine Bellahsen-Harrar. 2025. Exploring the Risks of Over-Reliance on Artificial Intelligence in Diagnostic Pathology. *Journal of Medical AI Research* (2025).
- [2] Aaron Chatterji, Tom Cunningham, David Deming, Zo'e Hitzig, Chen Ong, Canice Shan, and Kevin Wadman. 2025. *How People Use ChatGPT*. Technical Report 34255. National Bureau of Economic Research.
- [3] Rudrajit Choudhuri, Ambareesh Ramakrishnan, Amreeta Chatterjee, Bianca Trinkenreich, Igor Steinmacher, Marco Gerosa, and Anita Sarma. 2024. Insights from the Frontline: GenAI Utilization among Software Engineering Students. *arXiv preprint arXiv:2412.15624* (2024).
- [4] Aayushi Dangol. 2025. Beyond Users: Supporting Children in Interpreting, Resisting, and Collaborating with AI. In *Proceedings of the 24th Interaction Design and Children Conference (IDC '25)*. ACM, 1180–1184.
- [5] Kate Goddard, Abdul Roudsari, and Jeremy C. Wyatt. 2012. Automation Bias: A Systematic Review of Frequency, Effect Mediators, and Mitigators. *Journal of the American Medical Informatics Association* 19, 1 (2012), 121–127.
- [6] Andreas Johannssen et al. 2025. The Crucial Role of Explainable Artificial Intelligence (XAI) in Health Care Management Systems. *Healthcare Informatics* (2025).
- [7] P. K. Kahr, G. Rooks, M. C. Willemsen, and C. C. P. Snijders. 2024. Understanding Trust and Reliance Development in AI Advice. *ACM Transactions on Interactive Intelligent Systems* 14, 4 (2024).
- [8] Carina Manger, Annalena Vogl, Emely Rosbach, Carina Ziegler, and Andreas Riener. 2023. Automation’s Not Perfect but Neither Are We: Unveiling Illusionary Control and Automation Bias in Automated Driving. In *Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications Adjunct (AutomotiveUI '23)*. ACM.
- [9] Stanley Milgram. 1963. Behavioral Study of Obedience. *Journal of Abnormal and Social Psychology* 67, 4 (1963), 371–378.
- [10] Samir Passi and Mihaela Vorvoreanu. 2022. *Overreliance on AI: Literature Review*. Technical Report. Microsoft Research.
- [11] Gabriele Romeo and Davide Conti. 2025. Exploring Automation Bias in Human–AI Collaboration: A Review and Implications for Explainable AI. *AI & Society* (2025).
- [12] Beatriz Severes, Caroline Carreira, Ana Beatriz Vieira, Eduardo Gomes, João Tiago Aparício, and Inês Pereira. 2023. The Human Side of XAI: Bridging the Gap between AI and Non-Expert Audiences. In *Proceedings of the 41st ACM International Conference on Design of Communication (SIGDOC '23)*.
- [13] Mel Slater. 2006. A Virtual Reprise of the Stanley Milgram Obedience Experiments. *PLoS ONE* 1, 1 (2006), e39.
- [14] K. K. L. Wong, Y. Han, Y. Cai, W. Ouyang, H. Du, and C. Liu. 2025. From Trust in Automation to Trust in AI in Healthcare: A 30-Year Longitudinal Review and an Interdisciplinary Framework. *Bioengineering* 12, 10 (2025), 1070.
- [15] Yi Wu. 2023. Integrating Generative AI in Education: How ChatGPT Brings Challenges for Future Learning and Teaching. *Journal of Advanced Research in Education* 2, 4 (2023), 6–10.

A OVERFLOW FROM OTHER SECTIONS

Prompt to Frame the Authority and Neutral AI Advice

Authority Prompt. Act as a botany professor who is the supervisor of a botany intern. Use the voice of a calm, confident authority figure. The tone should be firm, precise, and fact-based. Use declarative and **imperative** sentences. Avoid all hedging or speculative language. Ensure you do not include words such as *maybe*, *perhaps*, or *it seems*. Reference logic, evidence, or institutional reasoning when justifying actions. Maintain emotional neutrality while projecting confidence, legitimacy, and intellectual authority. Use the 11th-grade reading level in your response.

Non-Authority Prompt. Act as a botany professor who is the supervisor of a botany intern. Use the voice of a calm,

confident figure. The tone should be firm, precise, and fact-based. Do not use declarative and **imperative** sentences. Use hedging or speculative language approximately 50% of the time, including words such as *maybe*, *perhaps*, or *it seems*. Reference logic, evidence, or institutional reasoning when justifying actions. Maintain emotional neutrality while projecting confidence, legitimacy, and intellect. Use the 11th-grade reading level in your response.

Table 2

2. AI_Correctness

Estimates

Measure: MEASURE_1					
AI_Correctness	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound	
1	3.800	.426	2.902	4.698	
2	6.322	.522	5.221	7.424	

Pairwise Comparisons

Measure: MEASURE_1					
(I) AI_Correctness	(J) AI_Correctness	Mean Difference (I-J)	Std. Error	Sig. ^a	95% Confidence Interval for Difference Lower Bound Upper Bound
1	2	-2.522 [*]	.441	<.001	-3.452 -1.592
2	1	2.522 [*]	.441	<.001	1.592 3.452

Based on estimated marginal means

^a. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a
Pillai's trace	.658	32.722 ^a	1.000	17.000	<.001	.658	32.722	1.000
Wilks' lambda	.342	32.722 ^a	1.000	17.000	<.001	.658	32.722	1.000
Hotelling's trace	1.925	32.722 ^a	1.000	17.000	<.001	.658	32.722	1.000
Roy's largest root	1.925	32.722 ^a	1.000	17.000	<.001	.658	32.722	1.000

Each F tests the multivariate effect of AI_Correctness. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

Tests of Within-Subjects Contrasts

Measure: MEASURE_1									
Source	AI_Correctness	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a
AI_Correctness	Linear	60.268	1	60.268	32.722	<.001	.658	32.722	1.000
AI_Correctness * Condition	Linear	.057	1	.057	.031	.862	.002	.031	.053
Error(AI_Correctness)	Linear	31.311	17	1.842					

a. Computed using alpha = .05

Tests of Between-Subjects Effects

Measure: MEASURE_1									
Transformed Variable: Average									
Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a	
Intercept	970.668	1	970.668	143.601	<.001	.894	143.601	1.000	
Condition	.984	1	.984	.146	.708	.008	.146	.065	
Error	114.911	17	6.759						

a. Computed using alpha = .05

Figure 1

Table 3

Estimated Marginal Means

1. Authority or nonauthority condition (2 groups)

Estimates

Measure: MEASURE_1					
Authority or nonauthority condition (2 groups)	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound	
authority	5.700	.230	5.215	6.185	
nonauthority	4.611	.242	4.100	5.123	

Pairwise Comparisons

Measure: MEASURE_1					
(I) Authority or nonauthority condition (2 groups)	(J) Authority or nonauthority condition (2 groups)	Mean Difference (I-J)	Std. Error	Sig. ^a	95% Confidence Interval for Difference Lower Bound Upper Bound
authority	nonauthority	1.089 [*]	.334	.005	.384 1.794
nonauthority	authority	-1.089 [*]	.334	.005	-1.794 -.384

Based on estimated marginal means

^a. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Univariate Tests

Measure: MEASURE_1								
Contrast	Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a
1	5.616	1	5.616	10.622	.005	.385	10.622	.867
Error	8.989	17	.529					

The F tests the effect of Authority or nonauthority condition (2 groups). This test is based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Computed using alpha = .05

2. AI_Correctness

Estimates

Measure: MEASURE_1					
AI_Correctness	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound	
1	8.894	.333	8.191	9.598	
2	1.417	.297	.789	2.044	

Pairwise Comparisons

Measure: MEASURE_1					
(I) AI_Correctness	(J) AI_Correctness	Mean Difference (I-J)	Std. Error	Sig. ^a	95% Confidence Interval for Difference Lower Bound Upper Bound
1	2	7.478 [*]	.536	<.001	6.346 8.609
2	1	-7.478 [*]	.536	<.001	-8.609 -6.346

Based on estimated marginal means

^a. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a
Pillai's trace	.920	194.459 ^a	1.000	17.000	<.001	.920	194.459	1.000
Wilks' lambda	.080	194.459 ^a	1.000	17.000	<.001	.920	194.459	1.000
Hotelling's trace	11.439	194.459 ^a	1.000	17.000	<.001	.920	194.459	1.000
Roy's largest root	11.439	194.459 ^a	1.000	17.000	<.001	.920	194.459	1.000

Each F tests the multivariate effect of AI_Correctness. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. Authority or nonauthority condition (2 groups) * AI_Correctness

Estimates

Measure: MEASURE_1					
Authority or nonauthority condition (2 groups)	AI_Correctness	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
authority	1	8.900	.459	7.932	9.868
authority	2	2.500	.409	1.636	3.364
nonauthority	1	8.889	.484	7.868	9.909
nonauthority	2	.333	.432	-.577	1.244

Pairwise Comparisons

Measure: MEASURE_1					
(I) Authority or nonauthority condition (2 groups)	(J) Authority or nonauthority condition (2 groups)	Mean Difference (I-J)	Std. Error	Sig. ^a	95% Confidence Interval for Difference Lower Bound Upper Bound
1	authority	.011	.667	.987	-1.395 1.418
1	nonauthority	-.011	.667	.987	-1.418 1.395
2	authority	2.167 [*]	.595	.002	.912 3.422
2	nonauthority	-2.167 [*]	.595	.002	-3.422 -.912

Based on estimated marginal means

^a. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Univariate Tests

Measure: MEASURE_1								
AI_Correctness	Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a
Contrast	.001	1	.001	.000	.987	.000	.000	.050
1	35.789	17	2.105					
2	22.237	1	22.237	13.264	.002	.438	13.264	.929
Error	28.500	17	1.676					

Each F tests the simple effects of Authority or nonauthority condition (2 groups) within each level combination of the other effects shown. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Computed using alpha = .05

Figure 2

Table 4

2. AI_Correctness

Estimates

Measure: MEASURE_1

AI_Correctness	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	3.694	.384	2.885	4.504
2	5.978	.525	4.871	7.084

Pairwise Comparisons

Measure: MEASURE_1

		Mean Difference (I- J)	Std. Error	Sig. ^b	95% Confidence Interval for Difference ^a	
(I) AI_Correctness	(J) AI_Correctness				Lower Bound	Upper Bound
1	2	-2.283 [*]	.462	<.001	-3.259	-1.308
2	1	2.283 [*]	.462	<.001	1.308	3.259

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^c
Pillai's trace	.589	24.373 ^a	1.000	17.000	<.001	.589	24.373	.996
Wilks' lambda	.411	24.373 ^a	1.000	17.000	<.001	.589	24.373	.996
Hotelling's trace	1.434	24.373 ^a	1.000	17.000	<.001	.589	24.373	.996
Roy's largest root	1.434	24.373 ^a	1.000	17.000	<.001	.589	24.373	.996

Each F tests the multivariate effect of AI_Correctness. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

4. Authority or nonauthority condition (2 groups) * AI_Correctness

Estimates

Measure: MEASURE_1

		95% Confidence Interval			
Authority or nonauthority condition (2 groups)	AI_Correctness	Mean	Std. Error	Lower Bound	Upper Bound
authority	1	3.100	.528	2.386	4.614
	2	5.400	.722	3.877	6.923
nonauthority	1	3.889	.557	2.715	5.063
	2	6.556	.761	4.950	8.161

Pairwise Comparisons

Measure: MEASURE_1

		Mean Difference (I- J)		95% Confidence Interval for Difference ^a			
Authority or nonauthority condition (2 groups)	(I) AI_Correctness	(J) AI_Correctness	Mean Difference (I- J)	Std. Error	Sig. ^b	Lower Bound	Upper Bound
authority	1	2	-1.900 [*]	.637	.008	-3.243	-.557
	2	1	1.900 [*]	.637	.008	-.557	3.243
nonauthority	1	2	-2.667 [*]	.671	<.001	-4.082	-1.251
	2	1	2.667 [*]	.671	<.001	1.251	4.082

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^c	
authority	Pillai's trace	.344	8.907 ^a	1.000	17.000	.008	.344	8.907	.803
	Wilks' lambda	.656	8.907 ^a	1.000	17.000	.008	.344	8.907	.803
	Hotelling's trace	.524	8.907 ^a	1.000	17.000	.008	.344	8.907	.803
	Roy's largest root	.524	8.907 ^a	1.000	17.000	.008	.344	8.907	.803
nonauthority	Pillai's trace	.482	15.791 ^a	1.000	17.000	<.001	.482	15.791	.963
	Wilks' lambda	.518	15.791 ^a	1.000	17.000	<.001	.482	15.791	.963
	Hotelling's trace	.929	15.791 ^a	1.000	17.000	<.001	.482	15.791	.963
	Roy's largest root	.929	15.791 ^a	1.000	17.000	<.001	.482	15.791	.963

Each F tests the multivariate simple effects of AI_Correctness within each level combination of the other effects shown. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

3. Authority or nonauthority condition (2 groups) * AI_Correctness

Estimates

Measure: MEASURE_1

		95% Confidence Interval			
Authority or nonauthority condition (2 groups)	AI_Correctness	Mean	Std. Error	Lower Bound	Upper Bound
authority	1	3.500	.528	2.386	4.614
	2	5.400	.722	3.877	6.923
nonauthority	1	3.889	.557	2.715	5.063
	2	6.556	.761	4.950	8.161

Pairwise Comparisons

Measure: MEASURE_1

		Mean Difference (I- J)		95% Confidence Interval for Difference ^a			
Authority or nonauthority condition (2 groups)	(I) Authority or nonauthority condition (2 groups)	(J) Authority or nonauthority condition (2 groups)	Mean Difference (I- J)	Std. Error	Sig. ^b	Lower Bound	Upper Bound
1	authority	nonauthority	-.389	.767	.619	-2.007	1.230
	nonauthority	authority	.389	.767	.619	-1.230	2.007
2	authority	nonauthority	-1.156	1.049	.286	-3.369	1.058
	nonauthority	authority	1.156	1.049	.286	-1.058	3.369

Based on estimated marginal means

a. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Figure 3

Table 5

3. Time

Estimates

Measure: MEASURE_1

		95% Confidence Interval			
Time	Mean	Std. Error	Lower Bound	Upper Bound	
1	2.979	.294	2.359	3.600	
2	4.444	.321	3.767	5.120	

Pairwise Comparisons

Measure: MEASURE_1

		Mean Difference (I- J)		95% Confidence Interval for Difference ^a		
(I) Time	(J) Time	Mean Difference (I- J)	Std. Error	Sig. ^b	Lower Bound	Upper Bound
1	2	-1.464 [*]	.269	<.001	-2.031	-.898
2	1	1.464 [*]	.269	<.001	.898	2.031

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

2. AI_Correctness

Estimates

Measure: MEASURE_1

		95% Confidence Interval			
AI_Correctness	Mean	Std. Error	Lower Bound	Upper Bound	
1	3.804	.276	3.223	4.385	
2	3.619	.284	3.020	4.218	

Pairwise Comparisons

Measure: MEASURE_1

		Mean Difference (I- J)		95% Confidence Interval for Difference ^a		
(I) AI_Correctness	(J) AI_Correctness	Mean Difference (I- J)	Std. Error	Sig. ^b	Lower Bound	Upper Bound
1	2	-.184 [*]	.080	.034	-.354	.354
2	1	.184 [*]	.080	.034	-.354	.354

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^c
Pillai's trace	.237	5.294 ^a	1.000	17.000	.034	.237	5.294	.583
Wilks' lambda	.763	5.294 ^a	1.000	17.000	.034	.237	5.294	.583
Hotelling's trace	.311	5.294 ^a	1.000	17.000	.034	.237	5.294	.583
Roy's largest root	.311	5.294 ^a	1.000	17.000	.034	.237	5.294	.583

Each F tests the multivariate effect of AI_Correctness. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

5. Authority or nonauthority condition (2 groups) * AI_Correctness

Estimates

Measure: MEASURE_1

		95% Confidence Interval			
Authority or nonauthority condition (2 groups)	AI_Correctness	Mean	Std. Error	Lower Bound	Upper Bound
authority	1	3.530	.379	2.730	4.330
	2	3.450	.391	2.626	4.274
nonauthority	1	4.078	.400	3.234	4.921
	2	3.789	.412	2.920	4.658

Pairwise Comparisons

Measure: MEASURE_1

		Mean Difference (I- J)		95% Confidence Interval for Difference ^a			
Authority or nonauthority condition (2 groups)	(I) AI_Correctness	(J) AI_Correctness	Mean Difference (I- J)	Std. Error	Sig. ^b	Lower Bound	Upper Bound
authority	1	2	-.080	.110	.478	-.313	.313
	2	1	.080	.110	.478	-.313	.313
nonauthority	1	2	-.289 [*]	.116	.024	-.534	-.043
	2	1	.289 [*]	.116	.024	-.043	.534

Based on estimated marginal means

*. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests

	Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^c	
authority	Pillai's trace	.030	.526 ^a	1.000	17.000	.478	.030	.526	.105
	Wilks' lambda	.970	.526 ^a	1.000	17.000	.478	.030	.526	.105
	Hotelling's trace	.031	.526 ^a	1.000	17.000	.478	.030	.526	.105
	Roy's largest root	.031	.526 ^a	1.000	17.000	.478	.030	.526	.105
nonauthority	Pillai's trace	.266	6.169 ^a	1.000	17.000	.024	.266	6.169	.649
	Wilks' lambda	.734	6.169 ^a	1.000	17.000	.024	.266	6.169	.649
	Hotelling's trace	.363	6.169 ^a	1.000	17.000	.024	.266	6.169	.649
	Roy's largest root	.363	6.169 ^a	1.000	17.000	.024	.266	6.169	.649

Each F tests the multivariate simple effects of AI_Correctness within each level combination of the other effects shown. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

b. Computed using alpha = .05

8. AI_Correctness * Time

Estimates							
Measure: MEASURE_1							
AI_Correctness	Time	Mean	Std. Error	95% Confidence Interval			
				Lower Bound	Upper Bound		
1	1	2.984	.293	2.365	3.603		
	2	4.623	.322	3.945	5.302		
2	1	2.974	.299	2.344	3.605		
	2	4.264	.329	3.571	4.958		

Pairwise Comparisons							
Measure: MEASURE_1							
Time	(I) AI_Correctness	(J) AI_Correctness	Mean Difference (I-J)	Std. Error	Sig. ^b	95% Confidence Interval for Difference ^a	
						Lower Bound	Upper Bound
1	1	2	.010	.068	.884	-.133	.153
	2	1	-.010	.068	.884	-.153	.133
2	1	2	.359 ^a	.109	.004	.130	.588
	2	1	-.359 ^a	.109	.004	-.588	-.130

Based on estimated marginal means

a. The mean difference is significant at the .05 level.

b. Adjustment for multiple comparisons: Least Significant Difference (equivalent to no adjustments).

Multivariate Tests									
Time		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power ^a
1	Pillai's trace	.001	.022 ^a	1.000	17.000	.884	.001	.022	.052
	Wilks' lambda	.999	.022 ^a	1.000	17.000	.884	.001	.022	.052
	Hotelling's trace	.001	.022 ^a	1.000	17.000	.884	.001	.022	.052
	Roy's largest root	.001	.022 ^a	1.000	17.000	.884	.001	.022	.052
	Pillai's trace	.391	10.919 ^a	1.000	17.000	.004	.391	10.919	.876
2	Wilks' lambda	.609	10.919 ^a	1.000	17.000	.004	.391	10.919	.876
	Hotelling's trace	.642	10.919 ^a	1.000	17.000	.004	.391	10.919	.876
	Roy's largest root	.642	10.919 ^a	1.000	17.000	.004	.391	10.919	.876

Each F tests the multivariate simple effects of AI_Correctness within each level combination of the other effects shown. These tests are based on the linearly independent pairwise comparisons among the estimated marginal means.

a. Exact statistic

Table 6

Independent t-test Results

T-Test

Group Statistics				
Authority Condition	N	Mean	Std. Deviation	Std. Error
Plaut,Confidence	10	3.10	1.449	.458
Nonauthority	9	3.22	1.392	.454

Independent Samples Test

Levene's Test for Equality of Variances			t-test for Equality of Means								
	F	Sig.	t	df	Significance One-tailed p	Two-Tailed p	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference Lower	Upper	
Plaut,Confidence	Equal variances assumed	.010	.922	-.193	17	.425	.850	-.122	.635	-1.462	1.217
	Equal variances not assumed			-.194	17.000	.424	.849	-.122	.631	-1.454	1.209

Independent Samples Effect Sizes

		95% Confidence Interval		
Standardized ^a	Point Estimate	Lower	Upper	
Plaut,Confidence	Cohen's d	-.382	-.088	.814
	Hedges' correction	-.447	-.084	.777
Class's skills		-.302	-.094	.993

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's skills uses the sample standard deviation of the control (i.e., the second) group.